

PH3202 (Intermediate EM)

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Class Test 2

Question 1 (5 marks)

Using Maxwell's equations, show that in vacuum, magnetic field satisfies the equation:

$$\mu_0\epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = \nabla^2 \vec{B}$$

Question 2 (5 marks)

Show that the general solution of the 1+1 dimensional wave equation:

$$\frac{1}{v_0^2} \frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = 0$$

can be expressed as:

$$f(t, x) = f_+(v_0 t + x) + f_-(v_0 t - x)$$